

SecureGate: Surface-Code Backend Validation and State-Preparation Fingerprints

Marya Alturki

September 25, 2026

Purpose. This report documents the work implemented in `01_backend_test.ipynb`. The notebook establishes a reproducible baseline for SecureGate by validating the simulation environment, constructing a rotated surface-code memory experiment, generating detector data under circuit-level noise, and characterizing the spacetime detector fingerprints associated with logical-state preparation. Its purpose is to provide a compact scientific roadmap so that the notebook can be followed in execution order without reconstructing the motivation of each cell.

1. Simulation environment and experiment parameters.

The notebook was executed in the `Python (Qiskit Project)` environment, using Python 3.12.14, Stim 1.16.0, `surface-sim` 0.11.0, and NumPy 2.5.2. The simulation parameters were

$$d = 3, \quad N_{\text{rounds}} = 5, \quad p = 10^{-3}, \quad N_{\text{shots}} = 10,000,$$

where d is the surface-code distance, N_{rounds} is the number of syndrome-extraction rounds, p is the physical error probability parameter used by the circuit-level noise model, and N_{shots} is the number of Monte Carlo samples.

A rotated surface-code layout was constructed with 9 data qubits and 17 total physical qubits, corresponding to 8 syndrome-extraction ancillas. The circuit-level noise model was configured using the physical error probability $p = 10^{-3}$, and detector definitions were generated directly from the surface-code layout.

2. Logical $|0\rangle_L$ memory experiment.

The first experiment initializes all data qubits in the computational basis, corresponding to the physical preparation

$$|0\rangle^{\otimes 9},$$

and uses the Z -basis memory configuration of `surface-sim`. The fault-tolerant Stim circuit generated by the notebook contains

$$17 \text{ physical qubits}, \quad 44 \text{ detectors}, \quad 1 \text{ logical observable}.$$

The detector sampler was then run for 10,000 shots, producing a binary matrix

$$D_Z \in \{0, 1\}^{10000 \times 44}.$$

Each row represents one noisy execution, while each column corresponds to one detector across the sampled executions.

For detector j , the notebook estimates the empirical detector-event probability as

$$\hat{P}(d_j = 1) = \frac{1}{N} \sum_{n=1}^N d_j^{(n)}.$$

The first ten estimated probabilities were

$$(0, 0, 0, 0, 0.0229, 0.0162, 0.0157, 0.0209, 0.0277, 0.0375),$$

and the overall detector-event probability across all detector entries and shots was

$$0.02683636.$$

3. Logical $|+\rangle_L$ memory experiment.

A second memory experiment was generated using the same physical layout, noise model, and number of syndrome-extraction rounds, but with the X -basis memory configuration. The resulting circuit again contains 17 physical qubits, 44 detectors, and one logical observable. Sampling produced

$$D_X \in \{0, 1\}^{10000 \times 44}.$$

The first ten empirical detector-event probabilities were

$$(0.0172, 0.0266, 0.0249, 0.0124, 0, 0, 0, 0, 0.0222, 0.0357),$$

with an overall detector-event probability of

$$0.02703636.$$

The notebook then compares the two detector-rate vectors $\{\hat{P}_X(d_j = 1)\}$ and $\{\hat{P}_Z(d_j = 1)\}$ detector-by-detector. The visualization provides the first direct empirical representation of the two logical-state preparation fingerprints.

4. Detector-wise fingerprint difference.

The detector-wise difference is defined in the notebook as

$$\Delta_j = \hat{P}_X(d_j = 1) - \hat{P}_Z(d_j = 1).$$

The mean absolute difference across all 44 detectors is

$$\frac{1}{44} \sum_{j=1}^{44} |\Delta_j| = 0.00509545,$$

while the maximum absolute difference is

$$\max_j |\Delta_j| = 0.0266,$$

occurring at detector $j = 1$.

The ten detectors with the largest observed absolute differences are

Detector	Z	X	\Delta
1	0.00000	0.02660	0.02660
2	0.00000	0.02490	0.02490
4	0.02290	0.00000	0.02290
7	0.02090	0.00000	0.02090
0	0.00000	0.01720	0.01720
5	0.01620	0.00000	0.01620
6	0.01570	0.00000	0.01570
3	0.00000	0.01240	0.01240
34	0.03570	0.04180	0.00610
8	0.02770	0.02220	0.00550

These values provide candidate detector locations where the two state-preparation conditions differ most strongly in this particular noisy simulation.

5. Spacetime structure of the detector fingerprint.

Detector indices were mapped to their spacetime coordinates

$$(x, y, t),$$

allowing the fingerprint difference to be examined across the decoding window rather than only as a one-dimensional detector-index plot. The notebook identifies six temporal coordinates:

$$t \in \{0, 1, 2, 3, 4, 4.5\}.$$

For every time slice, the detector-wise difference Δ_j is reported and visualized using a common scale. This preserves the spatial and temporal structure of the detector fingerprint and makes the sign of Δ_j explicit: positive values correspond to a higher empirical event rate for the X -basis preparation, while negative values correspond to a higher rate for the Z -basis preparation.

The largest early-window differences occur at $t = 0$, while later slices contain smaller but nonzero differences. For example, the largest observed differences in the $t = 0$ slice are

$$\Delta_1 = +0.0266, \quad \Delta_2 = +0.0249, \quad \Delta_4 = -0.0229, \quad \Delta_7 = -0.0209.$$

The spacetime visualization therefore shows that the distinction between the two preparations is structured rather than being represented only by a change in one global average.

6. Ancilla roles and stabilizer-family check.

The layout inspection identifies 9 data qubits,

$$(D1, \dots, D9),$$

and 8 ancillas divided into four X -type ancillas

$$(X1, X2, X3, X4)$$

and four Z -type ancillas

$$(Z1, Z2, Z3, Z4).$$

Their reported coordinates are

$$\begin{array}{c|cc} X\text{-type} & X1 & (0, 2) & X2 & (2, 4) \\ & X3 & (4, 2) & X4 & (6, 4) \end{array}$$

and

$$\begin{array}{c|cc} Z\text{-type} & Z1 & (2, 2) & Z2 & (2, 6) \\ & Z3 & (4, 0) & Z4 & (4, 4). \end{array}$$

For the initial detector layer, detectors 0–3 are grouped as the X -family and detectors 4–7 as the Z -family. The measured means are

	X -family	Z -family
$ 0\rangle_L$	0.00000	0.018925
$ +\rangle_L$	0.020275	0.00000

in the notebook output.

Thus, at the beginning of the decoding window, the two logical-state preparation conditions exhibit complementary behavior across the two stabilizer families. This provides a concrete baseline for understanding how state preparation can produce structured detector signatures before gate-level fingerprinting is introduced.

7. Role in the SecureGate workflow.

Notebook 01 establishes the simulation and detector-analysis baseline used by later notebooks. It verifies that the selected backend can generate noisy fault-tolerant surface-code circuits and that detector records can be represented as spacetime fingerprints with measurable, detector-dependent structure. The results here concern $|0\rangle_L$ versus $|+\rangle_L$ state preparation, not logical-gate inference. Consequently, the observed differences should be treated as a state-preparation baseline rather than as evidence of a gate-specific side channel.

Reference. Shukla et al., “Anticipating Decoder Side-channel Attacks in Fault-tolerant Quantum Computers,” arXiv:2607.12174 (2026).